# BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SUKKUR

INTERMEDIATE (11TH CLASS) EXAMINATION, 2019

# PHYSICS

Past Paper 2019 • Paper Code: B

Retyped faithfully from the original board paper — same questions, no changes.

## SECTION – C (MULTIPLE CHOICE QUESTIONS)

**Time Allowed:** 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** B

Note: Q.1: Choose the correct answer from the given options and write the same in your answer copy.

Q1. The term torque is synonymous with:

(a) Angular velocity
(b) Angular momentum
(c) Moment of inertia
(d) Moment of force

Q2. 0.1 horse power is equal to:

(a) 0.746 watt
(b) 7.46 watt
(c) 74.6 watt
(d) 746 watt

Q3. A converging lens behaves like a diverging lens, when an object is placed:

(a) At 2F
(b) Between F and 2F
(c) Beyond 2F
(d) Between Focus and optical centre

Q4. The image of distant object is seen through an astronomical telescope is:

(a) Erect
(b) Inverted
(c) Small in size
(d) None of these

Q5. A particle takes 0.2 sec to complete one revolution, its frequency is:

(a) 5 Hz
(b) 10 Hz
(c) 50 Hz
(d) 75 Hz

Q6. A lens whose power is 2 diopter, its focal length is:

(a) 5 cm
(b) 25 cm
(c) 50 cm
(d) 75 cm

Q7. Inertia of a body is measured in terms of its:

(a) Weight
(b) Mass
(c) Reaction
(d) Velocity

Q8. Angular velocity is directed parallel to:

(a) Tangential velocity
(b) Tangential acceleration
(c) Centripetal acceleration
(d) Axis of rotation

Q9. Waves in which the particles of medium undergoes the displacement parallel to the wave motion are called:

(a) Stationary waves
(b) Mechanical waves
(c) Longitudinal waves
(d) Transverse waves

Q10. Distance between two nearest points of wave having same phase is equal to:

(a) $\lambda/2$
(b) $\lambda/4$
(c) $\lambda$
(d) $2\lambda$

Q11. $\pi/6$ radians are equal to:

(a) 30°
(b) 60°
(c) 90°
(d) 120°

Q12. Speed of waves in air in term of wave length "λ" and time period "T" is:

(a) $\lambda T$
(b) $T/\lambda$
(c) $\lambda/T$
(d) $\lambda T$

Q13. A car travelling at a constant speed of 72 Km/h round the radius of curvature of 100m. What is its acceleration?

(a) 0.04 $m/s^2$
(b) 0.4 $m/s^2$
(c) 4 $m/s^2$
(d) 40 $m/s^2$

Q14. If the angular speed of a particle is "ω" its centripetal acceleration is:

(a) $\omega/r$
(b) $\omega r$
(c) $\omega^2 r$
(d) $\omega r^2$

Q15. Passage of sound wave through air is:

(a) Isobaric
(b) Isothermal
(c) Adiabatic
(d) Isochoric

Q16. The vivid colours observed on thin film is result of:

(a) Diffraction
(b) Interference
(c) Polarization
(d) Refraction

Q17. A force in its direction develops velocity of 100 m/sec. The power of the force is:

(a) 5F (b) 10 F

(c) 50F (d) 100 F

## SECTION – A (Short Answer Questions)

**Time Allowed:** 2 Hours and 40 Minutes | **Maximum Marks:** 40

Note: ATTEMPT ANY EIGHT (08) QUESTIONS. ALL QUESTIONS CARRY EQUAL MARKS.

Q1. State and Explain the Significant Figures.

Q2. Determine Unit vector perpendicular to the plane containing $\vec{A}$ & $\vec{B}$ when $\vec{A} = 2\hat{i} - 3\hat{j} - \hat{k}$ and $\vec{B} = \hat{i} + 4\hat{j} - 2\hat{k}$.

Q3. A ball is thrown vertically upward with speed of 10 m/sec. How high the ball will rise and how long it will remain in air?

Q4. Discuss for cases of special interest in elastic collision in one dimension.

Q5. Calculate the mass of earth and its average density.

Q6. Find value of “g” at a distance equal to radius of earth above the earth’s surface.

Q7. What is transverse wave? Discuss its important features.

Q8. What do you know about the angular momentum? Determine its unit and dimensions.

Q9. What are hyperopic and myopic defects of human eye and how are they corrected artificially?

Q10. What is Second Pendulum? Calculate the length of Second pendulum.

Q11. Drive an expression for Bragg’s Law by describing the diffraction of x-rays through crystals.

Q12. A convex lens of focal length 20 cm is used to form an erect image which is twice as large as the object. Find the position of object.

## SECTION – B (Long Answer Questions)

**Time Allowed:** 2 Hours and 40 Minutes | **Maximum Marks:** 28

Note: ATTEMPT ANY TWO (02) QUESTIONS. ALL QUESTIONS CARRY EQUAL MARKS.

Q13.

(a) [09] Define angular motion and angular velocity. Derive an expression for centripetal acceleration.

(b) [05] A car of weight 98N turns along a curved path of diameter 20 m with speed of 10 m/sec. Find centripetal force acting on the car.

Q14.

(a) [09] Calculate Kinetic energy and Potential energy of a particle executing simple harmonic motion and show that total energy of particle remains constant.

(b) [05] A body of mass 25 gm is attached to a spring of spring constant 0.4 $Nm^{-1}$. The body is displaced through 10 cm and released its velocity is found to be 0.4 m/sec at the end position. Calculate the total energy of the body.

Q15.

(a) [09] What is interference of light waves? Describe Young's double slit experiment and determine the position of bright and dark fringes.

(b) [05] In double slit experiment the separation of slits is 2mm and fringe spacing is 0.31mm at a distance of 100cm from the slits. Find the wave length of light in angstroms.